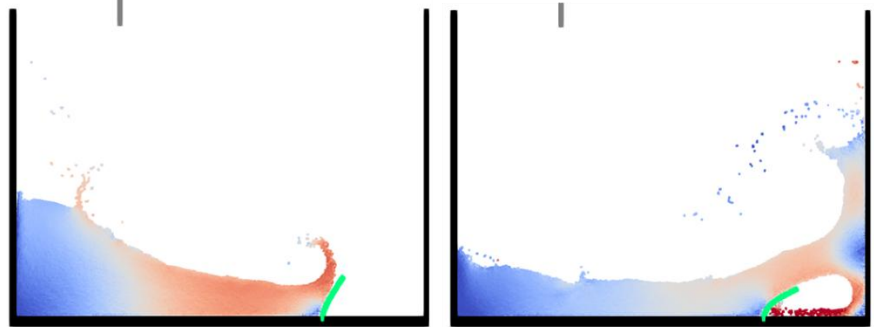


01_DAMBREAK

CASEDAMBREAK2D_FSI

- A 2-D reproduction of a dam break impacting an elastic plate.
- The setup is based on the experiment of [Liao et al. \(2015\)](#). However, the plate thickness and Young's modulus have been modified for this example.
- Realistic gate motion is also imposed to initiate the dam break.
- There are XMLs for both DBC and mDBC.

[Video](#)



Time: 0.30 s

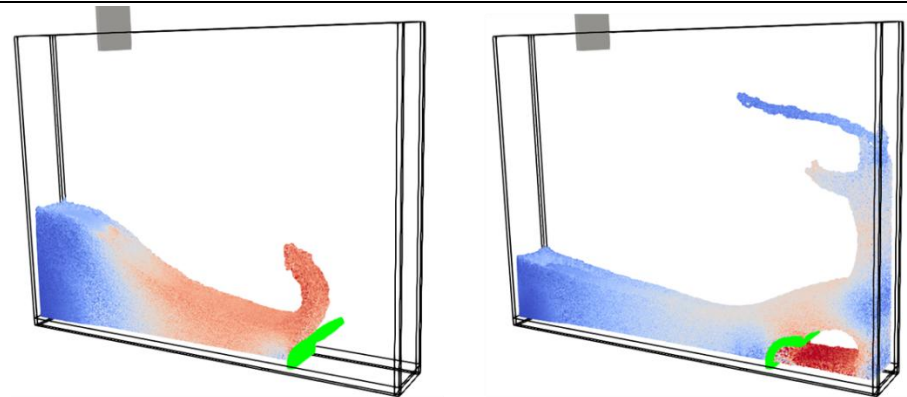
Time: 0.50 s

01_DAMBREAK

CASEDAMBREAK3D_FSI

- A 3-D reproduction of a dam break impacting an elastic plate.
- The setup is based on the experiment of [Liao et al. \(2015\)](#). However, the plate thickness, Young's modulus and spanwise (y-direction) length of the domain have been modified for this example.
- Realistic gate motion is also imposed to initiate the dam break.

[Video](#)



Time: 0.30 s

Time: 0.50 s

02_TUREKHON

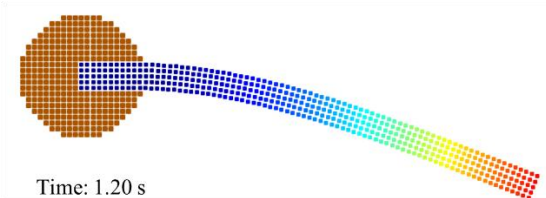
CASETUREKHONCSM3

- A reproduction of the CSM3 benchmark case from [Turek & Hron \(2006\)](#).
- The case consists of a 2-D clamped cantilever beam oscillating under gravity.

[Video](#)



Time: 0.00 s



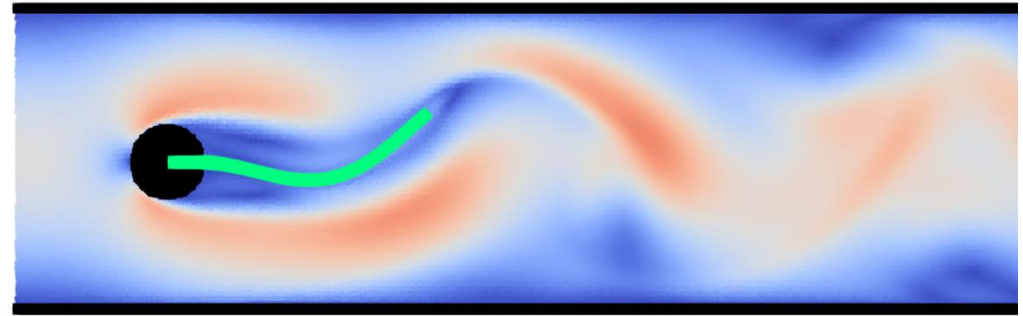
Time: 1.20 s

02_TUREKHON

CASETUREKHONFSI2

- A reproduction of the FSI2 benchmark case from [Turek & Hron \(2006\)](#).
- The case consists of a 2-D clamped cantilever beam attached to a 2-D cylinder embedded in a channel flow.
- The Reynolds number is 100, which leads to vortex shedding in the wake of the cylinder. These instabilities interact with the flexible beam, which ultimately leads to a self-sustaining flapping motion.
- There are XMLs for both DBC and mDBC.

[Video](#)



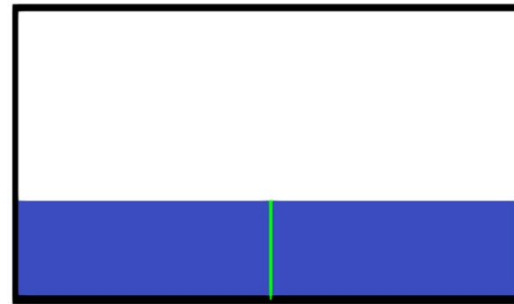
Time: 15.00 s

03_ROLLINGTANK

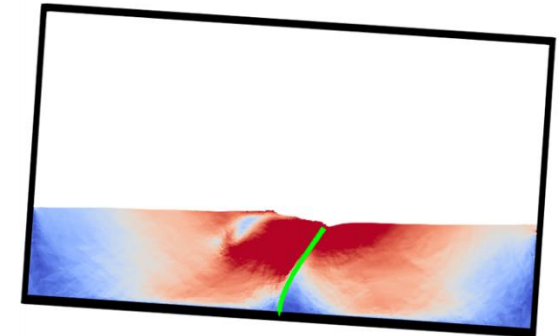
CASEROLLINGTANKDEEP

- A 2-D reproduction of the rolling tank experiment from [Idelsohn et al. \(2008\)](#), with an elastic beam attached to the bottom of the tank.
- The case consists of a standing beam in a deep oil solution so that the tip of the beam is flush with the top of the fluid level.
- The real tank motion from the experiment is imposed and the rolling frequency is set to match the critical sloshing frequency of the tank.
- The natural frequency of the flexible beam is also matched with the rolling frequency, leading to simple dynamics in the coupled fluid-structure interactions.

[Video](#)



Time: 0.00 s



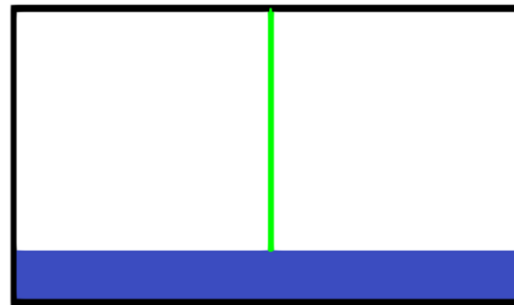
Time: 1.10 s

03_ROLLINGTANK

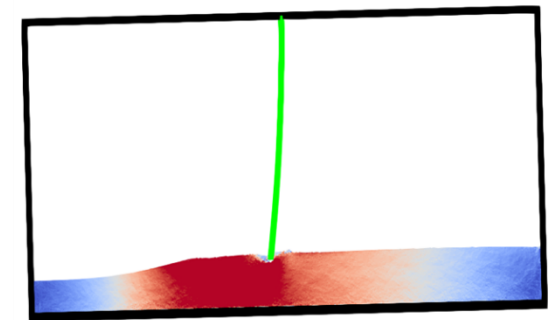
CASEROLLINGTANKHANGING

- A 2-D reproduction of the rolling tank experiment from [Idelsohn et al. \(2008\)](#), with an elastic beam attached to the top of the tank.
- The case consists of a hanging beam in a shallow water solution so that the tip of the beam is flush with the top of the fluid level.
- The real tank motion from the experiment is imposed and the rolling frequency is set to match the critical sloshing frequency of the tank.
- The natural frequency of the flexible beam is different from the rolling frequency, leading to chaotic dynamics in the motion of the beam as it is impacted by the sloshing waves generated by the tank motion.

[Video](#)



Time: 0.00 s



Time: 2.10 s